Multiplayer Connection— Test Report

Meijer Checkout Challenge

1. Purpose

This document independently verifies the reliability of the multiplayer connection across all three devices in the setup — the Host Tablet, the Client Tablet, and the PC (also acting as a client) — through multiple real-world scenarios, including role changes, disconnects/reconnects, leaderboard resets, date changes, and network topology variations.

2. Devices in Setup

- Tablet A / Tablet B — one acts as Host, the other as Client (roles are interchanged across tests).
- PC — always acts as a Client, and also drives the wall display.

3. System Behavior Reference

- Connection status for each of the 3 devices is checked automatically every 3 seconds.
- If connection is lost for any device, that device's status indicator turns RED within 3-10 seconds.
- For the PC, data automatically re-syncs once its connection is regained, no manual action required.
- For a Tablet, the connection auto-reconnects, but the tablet must be navigated back to the Home/First page for its data to re-sync, it does not resync automatically from wherever it was left.
- The PC-to-Router connection is supported over both WiFi and Ethernet; both configurations are tested below.

4. Network Topologies Tested

- Topology A: PC → Ethernet → Router → WiFi → Tablets
- Topology B: PC → WiFi → Router → WiFi → Tablets

5. Test Matrix

Each test should be iterated 3 times for confirmation of result. Fill in the status columns with the connection indicator observed on each of the 3 devices during/after the scenario (GREEN = connected, RED = disconnected, GREEN → RED → GREEN = disconnect/reconnect observed).

Test #	Scenario	Network Topology	Tablet A (Host)	Tablet B (Client)	PC (Client)	Result	Comments
1	Baseline connection — all 3 devices connect	PC→Ethernet→Router→WiFi→Tablets	Green	Green	Green	Successful	-
2	Interchange host/client roles between tablets, no leaderboard clear	PC→Ethernet→Router→WiFi→Tablets	Green → Red → Green	Green → Red → Green	Green → Red → Green	Successful	-
3	Disconnect host tablet, reconnect, then play a game	PC→Ethernet→Router→WiFi→Tablets	Green → Red → Green	Green → Red → Green	Green → Red → Green	Successful	-
4	Interchange host/client roles again, no leaderboard clear	PC→Ethernet→Router→WiFi→Tablets	Green → Red → Green	Green → Red → Green	Green → Red → Green	Successful	-
5	Disconnect client tablet, play 1 game on host, reconnect client	PC→Ethernet→Router→WiFi→Tablets	Green	Green → Red → Green	Green	Successful	-
6	Clear leaderboard on host (repeat with client in alt. iteration)	PC→Ethernet→Router→WiFi→Tablets	Green	Green	Green	Successful	Data is Reset
7	Disconnect host → clear leaderboard on client → reconnect host	PC→Ethernet→Router→WiFi→Tablets	Green → Red → Green	Green → Red → Green	Green → Red → Green	Successful	Data is Reserved
8	Disconnect client → clear leaderboard on host → reconnect client	PC→Ethernet→Router→WiFi→Tablets	Green	Green → Red → Green	Green	Successful	Data is Reset
9	Change dates ahead on all devices → clear leaderboard on host	PC→Ethernet→Router→WiFi→Tablets	Green	Green	Green	Successful	Data is Reset

Test #	Scenario	Network Topology	Tablet A (Host)	Tablet B (Client)	PC (Client)	Result	Comments
10	Repeat baseline connection test	PC→WiFi→Router→WiFi→Tablets	Green	Green	Green	Successful	-
11	Disconnect/reconnect host tablet mid-session	PC→WiFi→Router→WiFi→Tablets	Green → Red → Green	Green → Red → Green	Green → Red → Green	Successful	-
12	Disconnect/reconnect client tablet mid-session	PC→WiFi→Router→WiFi→Tablets	Green	Green → Red → Green	Green	Successful	-
13	Disconnect/reconnect PC (client) mid-session	PC→Ethernet→Router→WiFi→Tablets	Green	Green	Green→ Red → Green	Successful	-
14	Switch Off Router mid-session (Wifi Connection)	PC→Ethernet→Router→WiFi→Tablets	Green	Green → Red → Green	Green → Red	Partially Successful	Client tablets reconnect but PC unable to reconnect without restarting the application on PC
15	Switch Off Router mid-session (Ethernet Connection)	PC→WiFi→Router→WiFi→Tablets	Green	Green → Red → Green	Green → Red	Partially Successful	Client tablets reconnect but PC unable to reconnect without restarting the application on PC

6. Notes

- Pay particular attention to Tests 7 and 8 — these clear the leaderboard on one device while the other is disconnected, and may show differing reset behavior depending on which device the leaderboard is cleared on.
- Tests 10-13 cover WiFi-only PC connection and PC disconnect/reconnect scenarios.